

The Random Effects Model - Components of Variance

In some situations the factor levels (treatments) are not of intrinsic interest in themselves, and the factor (treatment) has a large number of possible levels. If the experimenter randomly selects a of these levels from the population of factor levels, then we say that the factor is random. In this case inferences are made about the entire population of factor levels. For the analysis we present below to be valid one has to assume that the population of factor levels is large enough to be considered infinite. For an analysis of a random effects model with a finite population of factor levels see: Searle and Fawcett (1970), *Biometrics* **26**, 243-254. Examples of random factors are: days, operators, workers, machines, batches of material, boxes of shipped items.

Example: Apex Enterprises, a company that builds roadside restaurants carrying one of several promoted trade names, leases franchises to individuals to operate the restaurants, and provide management services. This company employs a large number of personnel officers who interview applicants for jobs in the restaurants. At the end of the interview, the personnel officer assigns a subjective rating between 0 and 100 to indicate the applicants potential value on the job. Suppose now that five personnel officers were selected at random and each was assigned four candidates at random from a pool of candidates

that have the same potential (determined from a preliminary screening). In this case the company is not interested to compare the average ratings for the five personnel officers who happened to be selected for this study, but rather to answer some questions related to the entire population of personnel officers. Questions of interest might include: How large is the variation in ratings between all personnel officers? How this variation relates to the total variation between the ratings?

The model for this experiment looks similar to the fixed effects model:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

with the exception that τ_i are assumed to be independent and identically distributed random variables. The assumptions are:

For $i=1, \dots, a$ and $j=1, \dots, n_i$

1. μ is a fixed constant
2. τ_i are iid $N(0, \sigma_\tau^2)$
3. ε_{ij} are iid $N(0, \sigma_\varepsilon^2)$
4. τ_i and ε_{ij} are independent of each other

The variances σ_ε^2 and σ_τ^2 are called *variance components*. It follows from the assumptions that

$$\text{Var}(Y_{ij}) = \sigma_e^2 + \sigma_\tau^2.$$

The random effects model that was presented here differs from the fixed effects model in the following features:

1. $E(Y_{ij}) = \mu$ for any i and j
2. $Y_{i,j1}$ and $Y_{i,j2}$ are not independent of each other.

The dependence in the random effects model is formulated in a concept, useful in genetics and introduced by R. A. Fisher (1925, Ch. 7), called the *interclass correlation coefficient*:

$$\begin{aligned}\rho &= E[(Y_{i,j1} - \mu)(Y_{i,j2} - \mu)] / \text{Var}(Y_{ij}) \\ &= E[(\tau_i + \varepsilon_{i,j1})(\tau_i + \varepsilon_{i,j2})] / \text{Var}(Y_{ij}) \\ &= E(\tau_i^2) / \text{Var}(Y_{ij}) = \sigma_\tau^2 / \sigma_e^2 + \sigma_\tau^2.\end{aligned}$$

This measure is actually the correlation coefficient between any two observations for the same factor level, regardless of the actual level. It is also the proportion of the variation that is due to the factor levels.

Analysis of the Random Effects Model

Testing hypotheses about individual treatment means is meaningless, so instead we test the hypotheses:

$$H_0: \sigma_{\tau}^2 = 0 \text{ vs } H_1: \sigma_{\tau}^2 > 0.$$

The sum of squares identity

$$SS_{\text{total}} = SS_{\text{treatment}} + SS_{\text{error}}$$

is still valid and plays a major role in deriving the test statistic.

If $\sigma_{\tau}^2 = 0$, all treatments are identical; but if $\sigma_{\tau}^2 > 0$, variability exists between treatments.

As in the fixed effects model,

$$SS_{\text{error}} / \sigma_e^2 \sim \chi^2_{N-a},$$

and under the H_0 ,

$$SS_{\text{treatment}} / \sigma_e^2 \sim \chi^2_{a-1}.$$

Both random variables are independent and therefore under the H_0 the ratio

$$F = \frac{SS_{\text{treatment}} / (a-1)}{SS_{\text{error}} / (N-a)} = \frac{MS_{\text{treatment}}}{MS_{\text{error}}}$$

has an F distribution with $a-1$ and $N-a$ degrees of freedom.

Moreover one can show that:

$$E(MS_{\text{treatment}}) = \sigma_e^2 + n_0 \sigma_\tau^2$$

and

$$E(MS_{\text{error}}) = \sigma_e^2,$$

where

$$n_0 = \{(\sum n_i) - (\sum n_i^2) / (\sum n_i)\} / (a-1).$$

If $n_i = n$ (balanced design) then $n_0 = n$.

From the expected mean squares, we see that under H_0 both the numerator and the denominator of the test statistic are unbiased estimators of σ_e^2 , whereas under H_1 the expected value of the numerator is greater than the expected value of the denominator.

Therefore, we should reject the null hypothesis for large values of F . This implies an upper-tail, one-tail critical region and we

reject H_0 if $F > F_{\alpha; a-1, N-a}$.

Example: Ratings by Five Personnel Officers-Apex Enterprises

Officer j	Candidate i				
	1	2	3	4	
1	76	64	85	75	$\bar{Y}_{1.} = 75$
2	58	75	81	66	$\bar{Y}_{2.} = 70$
3	49	63	62	46	$\bar{Y}_{3.} = 55$
4	74	71	85	90	$\bar{Y}_{4.} = 80$
5	66	74	81	79	$\bar{Y}_{5.} = 71$
Mean					$\bar{Y}_{..} = 71$

ANOVA Table for Apex Enterprises

Source of Variation	SS	df	MS	E(MS)	F
Personnel Officers	1,480	4	370	$\sigma_e^2 + 4\sigma_\tau^2$	4.89
Error	1,134	15	75.6	σ_e^2	
Total	2,614	19			

Post F Test Estimation Procedures for a balanced design

1. Estimating μ
2. Estimating the components of variance

1. To estimate μ we use $\bar{Y}_{..}$.

Moreover, $\bar{Y}_{..}$ has a normal distribution with mean = μ and variance = $(\sigma_e^2 + n\sigma_\tau^2)/N$. An unbiased estimator of

this variance is given by $MS_{\text{treatment}} / N$.

Therefore, a $(1-\alpha)100\%$ confidence interval for μ is given by

$$\bar{Y}_{..} \pm t_{1-\alpha/2; n-1} [MS_{\text{treatment}} / N]^{.5}$$